Year 8 Earth and Space Sciences Revision Guide

Content

Learning intention			
Identify the 3 different types of rocks (sedimentary, igneous and metamorphic)			
Describe the characteristics, formation and examples of igneous rocks			
Describe the characteristics, formation and examples of sedimentary rocks			
Describe the characteristics, formation and examples of metamorphic rocks			
Identify minerals as the substances that rocks are made of			
Describe the changes in rocks and buildings over time			
Identify acidic rain as a cause of chemical weathering			
Explain how water absorbed by rocks expands on freezing and that the forces arising from expansion and contraction are great enough to break off pieces of rock			
Describe how temperature (onion-skin weathering) can result in rock fragmentation			
Explain that roots growing through and burrowing animals can contribute to biological weathering			
State that rock fragments can be transported by flowing water and the wind and that the sediments settle to form layers – remaking rock			
Describe the continuous process of the rock cycle			
Recognise that some rocks and minerals, such as ores, provide valuable resources			
Understand the different processes involved in the mining of metals			

Definitions:

- Rock
- Mineral
- Crystal
- Igneous rocks
- Intrusive rock
- Extrusive rock
- Sedimentary rock
- Erosion
- Sediments
- Fossils
- Layering
- Floodplains
- Coal
- Metamorphic rocks
- Parent rock
- Rock cycle
- Mineral ore
- Mining
- Environmental impact statement
- Rehabilitation
- Overburden
- Open cut mining
- Underground mining

Questions:

1) For each of the three different types of rocks, describe their characteristics, how they are formed, and some examples of them.

	Igneous	Sedimentary	Metamorphic
How it is formed			
Examples			
Does it have layers?			
Is it porous?			
Level of hardness?			
Does it have grains or crystals?			
The level of effect of weathering			
Diagram			

1. Write a description for each type of weathering

Weathering		
Chemical	Rain water	
	Acid rain	
Biological	Animals	
	Plants	
	Humans	
Physical	Temperature change	
	Wind, rain and waves	
	Freeze-thaw process	
	Exfoliation	

2) What are the similarities and differences between an intrusive rock, and an extrusive rock?

3) Some igneous rocks have pores in them. Are these rocks intrusive or extrusive? Why are there pores in the rock?

4) What are the major differences between igneous rocks and sedimentary rocks?

5) Describe the process in which sedimentary rocks are formed.

6) Why is running water an important part of the formation of sedimentary rocks?

7) What are the two types of materials can form sedimentary rocks?

8) How are igneous and sedimentary rocks transformed into metamorphic rocks?

9) What are minerals?

10) What daughter rocks do Limestone and Sandstone form under high heat?

11) What daughter rock does Shale form under high pressure?

12) What are the different types of weathering?

13) When walking through a park, you notice a large bed of rock has started to crack from it's underside. A large tree is nearby. What type of weathering is this? Explain how it occurs.

14) A large rock face is situated in an area which, over the course of the year, experiences an average temperature of around -5 degrees in winter, and 30 degrees in summer. Explain why there are deep cracks in the rock face. HINT: What does an average temperature mean?

15) What type of rocks are most susceptible to weathering?

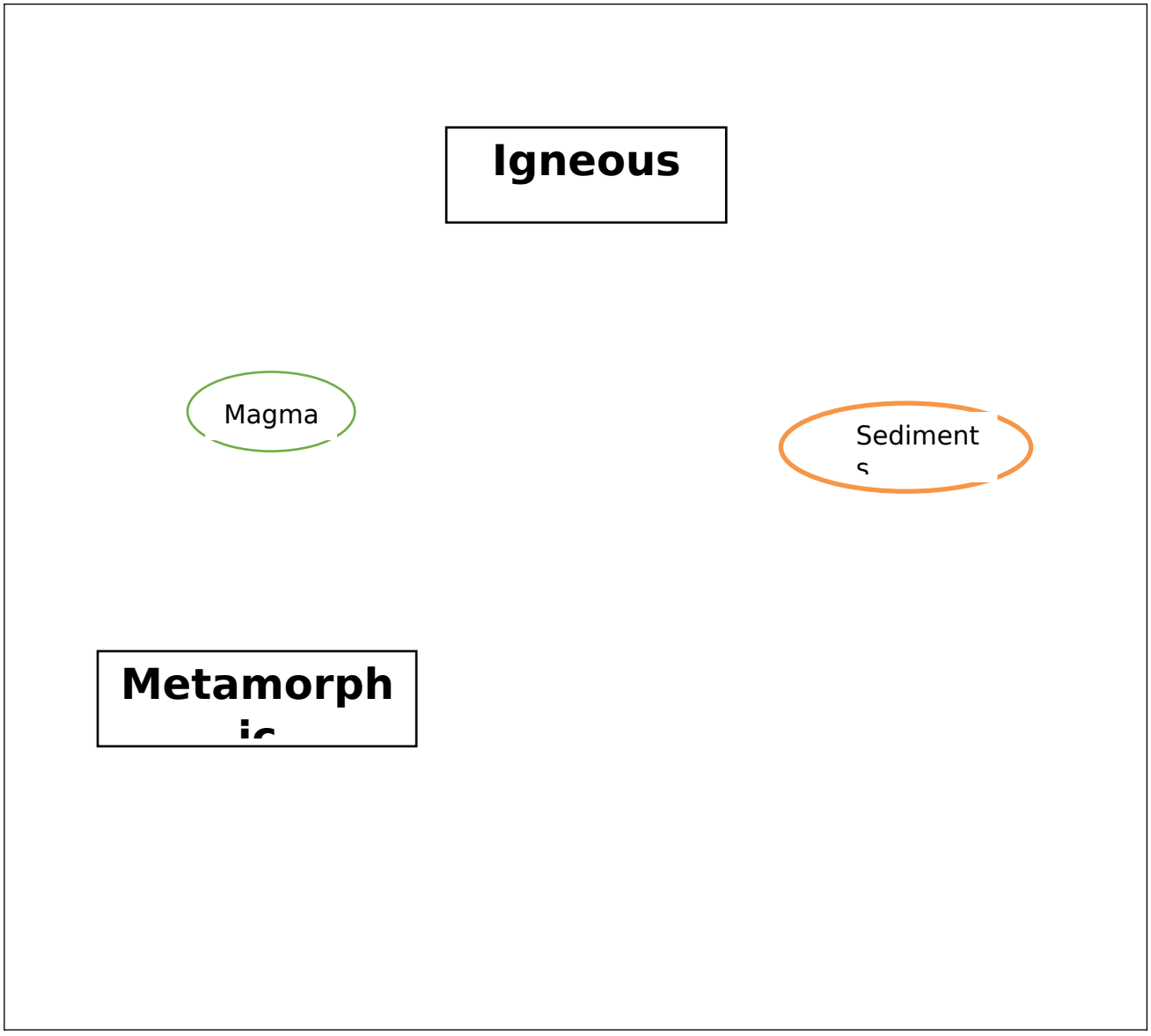
16) Explain the difference between weathering and erosion (use definitions and an example).

17) What are the two main types of mines?

18) What is required for a company to start mining?

19) How do mining companies get metal from ore?

20) Draw a labelled diagram of the rock cycle. Some has been done for you.



21) Label the following examples of rock as either igneous, sedimentary or metamorphic

a. Limestone:_____
b. Slate:_____
c. Conglomerate:_____
d. Pumice:_____
e. Rock salt:_____
f. Marble:_____
g. Granite:_____
h. Basalt:_____
i. Coal:_____
j. Shale:_____

Textbook references

Science Quest 8:

Section 9.1 Rocks and minerals (pg 278)

Section 9.2 Hot Rocks (pg 280)

Section 9.3 Sedimentary rocks (pg 283)

Section 9.4 Stability and change: Metamorphic rocks (pg 288)

Section 9.5 Mining for metals (pg 291)

Looking back (pg 304) Questions 1 – 16.